

Ruixuan Zhang

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EMPLOYMENT

University of Utah, Department of Mathematics
Postdoctoral Research Associate, C.R. Wiley Instructor

July 2024- Present

EDUCATION

University of Kansas

Aug. 2019- May 2024

PhD, Department of Mathematics

Dissertation: Conditional Limit Theorems for the KPZ Fixed Point with a Large Height.

Advisor: Zhipeng Liu

Indiana University, Indianapolis

Aug. 2017 - May 2019

PhD Student in Statistics, Department of Mathematical Sciences

Supervisor: Pavel Bleher

Ningbo University

Sep. 2013 - June 2017

Bachelor of Management, Accounting, ACCA Class, School of Business

Bachelor of Sciences, Mathematics, School of Sciences

RESEARCH INTERESTS

Kardar-Parisi-Zhang universality, integrable stochastic systems, random matrix theory and probabilistic models from physics.

PREPRINTS AND PUBLICATIONS

- *Coalescence and fluctuations of O'Connell–Yor polymers via the free-energy correlation profile*, with Firas Rassoul-Agha, Xiao Shen, Guangqu Zheng, *preprint*.
- *An upper tail field of the KPZ fixed point*, with Zhipeng Liu, *Communications in Mathematical Physics*, Volume 406, 198 (2025).
- *Edgeworth-type expansion for the one-point distribution of the KPZ fixed point with a large height at a prior location*, with Ron Nissim, *Ann Inst Henri Poincaré (B) Probab Stat*, 62(2), 2026.
- *Bayesian inference for thick-tailed stochastic volatility model- Empirical Study of Shanghai Composite Index*, with Wenli Huang, *Journal of Quantitative Economics*, Vol. 34 - Issue 4.

TEACHING EXPERIENCE

University of Utah

2024–present

MATH 3070, Applied Statistics I (Fall 2025, Fall 2026); MATH 1070, Introduction to Statistical Inference (Spring, Summer 2026); MATH 1030, Introduction to Quantitative Reasoning (Spring 2026); MATH 5090, Statistical Inference II (Spring 2025); MATH 3080, Applied Statistics II (Spring 2025); MATH 5080, Statistical Inference I (Fall 2024).

University of Kansas

2019–2024

Instructor: MATH 101, Algebra I (Fall 2019, Spring 2020); MATH 104, Pre-Calculus (Fall 2022); MATH 115, Calculus I (Fall 2020, Fall 2021); MATH 116, Calculus II (Spring 2021). Lab Instructor: MATH 126 (Spring 2023); MATH 127, Calculus III (Spring 2022, Spring 2024).

Indiana University

Indianapolis 2018

Instructor: MATH 165, Analytic Geometry and Calculus I (Fall 2018). Teaching Assistant: STAT 417, Statistical Theory, and STAT 350, Introduction to Statistics (Spring 2018).

RESEARCH TALKS

2026/03	Seminar on Stochastic Processes, Union College — <i>The coalescence and exit time of O’Connell–Yor polymers via Gaussian integration by parts.</i>
2025/09	Stochastics/Discrete Analysis, North Carolina State University — <i>An upper tail field of the KPZ fixed point.</i>
2025/09	Statistics Seminar, Indiana University, Indianapolis — <i>An upper tail field of the KPZ fixed point.</i>
2025/06	Workshop on Stochastic Interacting Particle Systems and Random Matrices, Rényi Institute, Budapest.
2025/03	Probability Seminar, Northwestern University — <i>An upper tail field of the KPZ fixed point.</i>
2024/02	Probability Seminar, University of Utah — <i>Conditional limit theorem of the KPZ fixed point with a large height.</i>
2023/11	22nd Northeast Probability Seminar, New York University — <i>A conditional limit theorem of the KPZ fixed point.</i>
2023/10	Probability Seminar, Texas Tech University — <i>Some exact formulas of the KPZ fixed point and its application.</i>
2023/03	Probability Seminar, University of Rochester — <i>Conditional limits of the KPZ fixed point with a large height at one location.</i>
2022/12	3rd Kansas Math Graduate Student Conference — <i>Asymptotics of the one-point distribution of the KPZ fixed point conditioned on a large height at an earlier point.</i>
2022/11	21st Northeast Probability Seminar, Columbia University — <i>Conditional limits of the KPZ fixed point with height tending to infinity at one location.</i>
2022/04	2nd Kansas Math Graduate Student Conference, Kansas State University — <i>Right tail of the two-time distribution of the KPZ fixed point.</i>
2021/12	Frontier Probability Days, University of Nevada — <i>Right tail of the two-time distribution of the KPZ fixed point.</i>

HONORS AND AWARDS

Outstanding Postdoc Award	2025
Department of Mathematics, University of Utah	
Paul F. Conrad Graduate Scholarship	2023
Department of Mathematics, University of Kansas	
Graduate Student Summer Scholarships	2023
Department of Mathematics, University of Kansas	
Graduate Student Travel Scholarships	2022
Department of Mathematics, University of Kansas	
Outstanding Beginning Graduate Student Awards	2018
School of Sciences, Indiana University Indianapolis	
Xinghao Jiang Research Scholarships	2015
School of Business, Ningbo University	